

Parallelize Graph Calls

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If we want to parallelize the extraction step, we'll need a robust deduplication approach. This messy doc lays out a naive, worst case approach in which we have to copmontentize our graph into disconnected sub-graphs each of approximately equal size. We then need to check each sub-graph for duplicates and then check each pair of sub-graphs for duplicates. We then need to repeat this process until we have no duplicates. This is a worst case scenario, but it gives us a starting point for thinking about the problem.

1 Graph and Tokens Definitions

Call our graph, G with nodes, $\mathcal{N}$ and edges, $\mathcal{E}$:

$$G = \{\mathcal{N}, \mathcal{E}\}$$

$$\mathcal{N} = \{n \in \text{nodes}\} \quad \mathcal{E} = \{e \in \text{edges}\}$$

$$\text{size}(G) = |\mathcal{N}|$$

Given a model, M , our token limit is T_M . Given a node, we count the tokens needed to represent it to the LLM (name and description) using a token function, $\mathcal{T}_M$.

$\mathcal{T}_M(\mathcal{N})$ is approximately normally distributed about its mean on $\mathbb{N}$ (for nodes)

$$\mathcal{T}_M(\mathcal{N}) \sim \mathcal{N}(\mu_{\mathcal{N}}, \sigma_{\mathcal{N}})$$

$\therefore$ we can include $\frac{T_M}{\mu_{\mathcal{N}}}$ total nodes; call this the “Model Graph Capacity”, C_M .

Ex: For Anthropic, the token limit is $T_A = 200k$

Assume approximately 50 tokens per node, $\mu_{\mathcal{N}} = 50$, implying $C_M = 4000$.

2 Async Deduplication

If we go async we need to de-dupe. Let's look at a concrete example:

$$\text{say } \text{size}(G) = 16,000$$

$$G = \bigoplus_{i=1}^4 g_i$$

To de-dupe we need to:

1. check each g_i individually
2. check each pair of components $g_i \oplus g_j$, note that due to the fact that components are disconnected, we have:

$$\text{size}(g_i \oplus g_j) = \text{size}(g_i) + \text{size}(g_j)$$

We need to subdivide any components g_i that were not reduced by at least a factor of 2 in step 1. Assume the worst case in which we don't see this reduction in any of our components, g_i . Therefore, we subdivide each g_i into g_{i1} and g_{i2} and perform pairwise graph component deduplication:

So we get to:

$$|\{g_i\}| = \frac{\text{size}(G)}{C_n}$$

We reduce further by at least a factor of 2 in step 2, so we have:

$$|\{g_i g_j\}| = 2 \frac{\text{size}(G)}{C_n}$$

Putting this all together, we count the total individual calls in step (1) above, as well as all the pairwise compares to give:

$$\begin{aligned} & |\{g_i\}| + |\{g_i g_j\}|(|g_i| - 1) \\ &= \frac{\text{size}(G)}{C_n} + \frac{2\text{size}(G)}{C_n^2} \left(\frac{\text{size}(G)}{C_n} - 1 \right) \\ &= 2 \left(\frac{\text{size}(G)}{C_n} \right)^2 \end{aligned}$$

So for our quantities,

$$\text{size}(G) = 16k$$

$$C_n = 4k$$

$$\# \text{ LLM calls} = 2 \left(\frac{16000}{4000} \right)^2 = 32$$